Chapter - Four

Theory of Cost

Topics discussed………..

- **Cost of production**
- **Cost function**
- **Short run and long run cost function**
- **TC, TFC, TVC**
- **AC, AFC, AVC**
- **MC**
- **Economies of Scale**

- ***COST OF PRODUCTION:***

- Cost of production refers to the total cost incurred by a business to produce a specific quantity of a product or offer a service.

- In economics, the cost of production is defined as the expenditures incurred to obtain the factors of production such as land, labor, and capital, that are needed in the production process of a product.

- Cost Function

- A cost function is a function of input prices and output quantity whose value is the cost of making that output given those input prices. Cost functions are derived functions. They are derived from the production function, which describes the available efficient methods of production at any one time. Cost is a function of output,

- C = f(X), ceteris paribus.
-

Economic theory distinguishes between short-run costs and long-run costs.

- ***Short run cost function***

- Short-run costs are the costs over a period during which some factors of production (usually capital equipment and management) are fixed.
- The short-run costs are the costs at which the firm operates in any one period.
-
- the short-run cost function
  - $C = f\,(X, T, P_f\,, \bar{k})$
- Where,
- C = total cost
- X= output
- T = technology
- P_f = prices of factors
- $\bar{k}$ = fixed factor(s)
-

- ***<u>Long-run cost function</u>***

- The long-run costs are the costs over a period long enough to permit the change of all factors of production. Such as planning cost. In the long run all factors become variable.
- Symbolically we may write the long-run cost function as

$$C = f\,(X, T, P_f)$$

- Where,
- C = total cost
- X= output
- T= technology
- P_f = prices of factors

Total Fixed Cost

-
- Fixed costs are the cost which are incurred in the fixed factors of production. More simply, fixed cost are the costs that don't changes with the quantity of output produced. Fixed costs are expenses that must be paid even if the firm produces zero output. They consist of items such as rent for factory or office space, interest payments on debts, salaries of tenured faculty and so forth.
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- *For example*; if you want to open a burger restaurant, you will need to pay rent for your location. Let's TK. 1000 per month. This is fixed cost because it does not matter how many burgers you sell; you will still have to pay the rent. Similarly, you have to pay your waitress' salary, regardless of the number of burgers she serves. If she/he earns TK. 1000 per month. Then your fixed costs add up to Tk. 2000 per month.

***Total Variable* cost**

•Variable costs are the cost which are incurred in the variable factors of production. More simply, Variable costs are the costs that changes with the quantity of output produced. That is, they usually increase as output increases and vice-versa. Unlike fixed costs, variable costs are not incurred if there is no production. They consist of item such as raw materials, wage and so forth.

•*For example,* in case of burger restaurant, the costs of meat, burger buns, lettuce, and BBQ sauce would be considered variable costs. Let’s assume all ingredients add up to TK. 10 per burger. If you sell 20 burgers and your variable costs are the costs of the ingredients, your total variable costs result in TK. 200. By contrast, if you sell 200 burgers, your total variable costs add up to TK. 2000. If you don’t sell any burgers at all, your total variable cost is zero.

Total cost

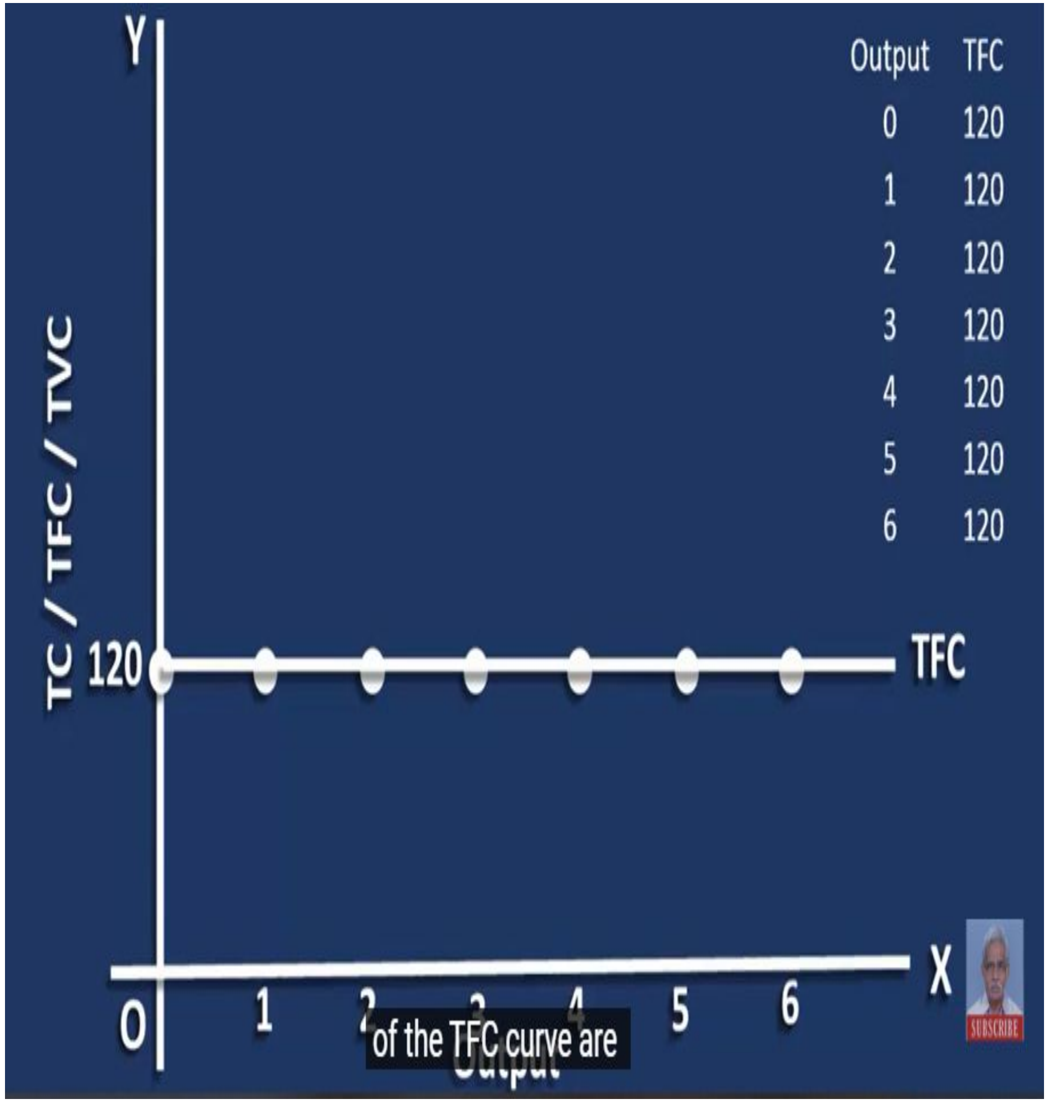
- Total cost describes the sum of total fixed cost and variable cost. It incurred all the costs that are incurred during the production process. Again, let’s say you managed to sell 200 burgers in your first month. In this case, your total cost of running your burger restaurant add up to TK. 4000 (i.e., Tk. 2000 Fixed cost + TK. 2000 variable cost).

Output	TFC
0	120
1	120
2	120
3	120
4	120
5	120
6	120

TC =TFC + TVC

Output	TVC
0	0
1	50
2	90
3	120
4	150
5	190
6	270

Output	TFC	TVC	TC
0	120	0	120
1	120	50	170
2	120	90	210
3	120	120	240
4	120	150	270
5	120	190	310
6	120	270	390


Y
Output TFC
0 120
1 120
2 120
3 120
4 120
5 120
6 120
TC / TFC / TVC
120
TFC
X
O
1
2
3
4
5
6
of the TFC curve are
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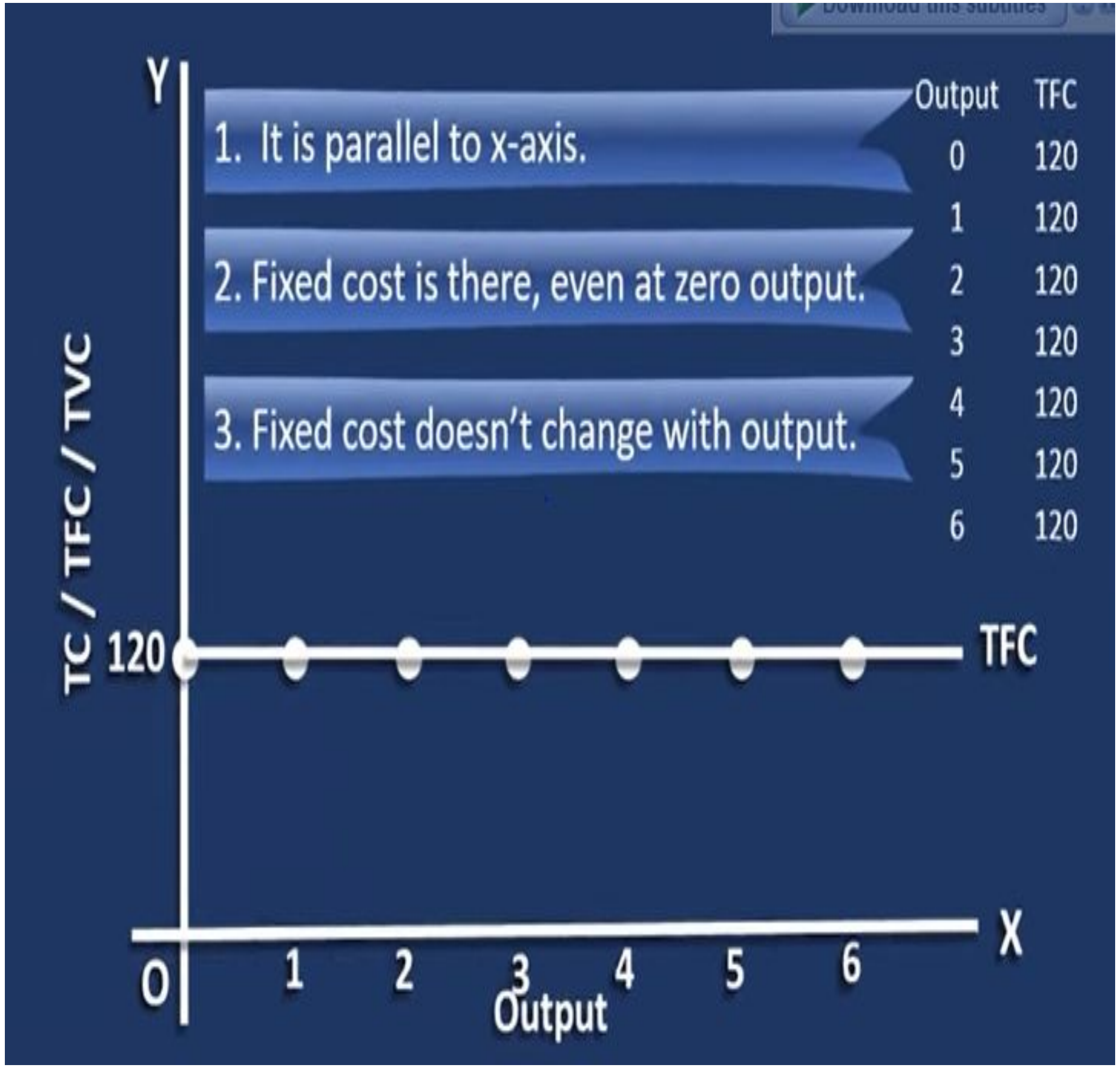
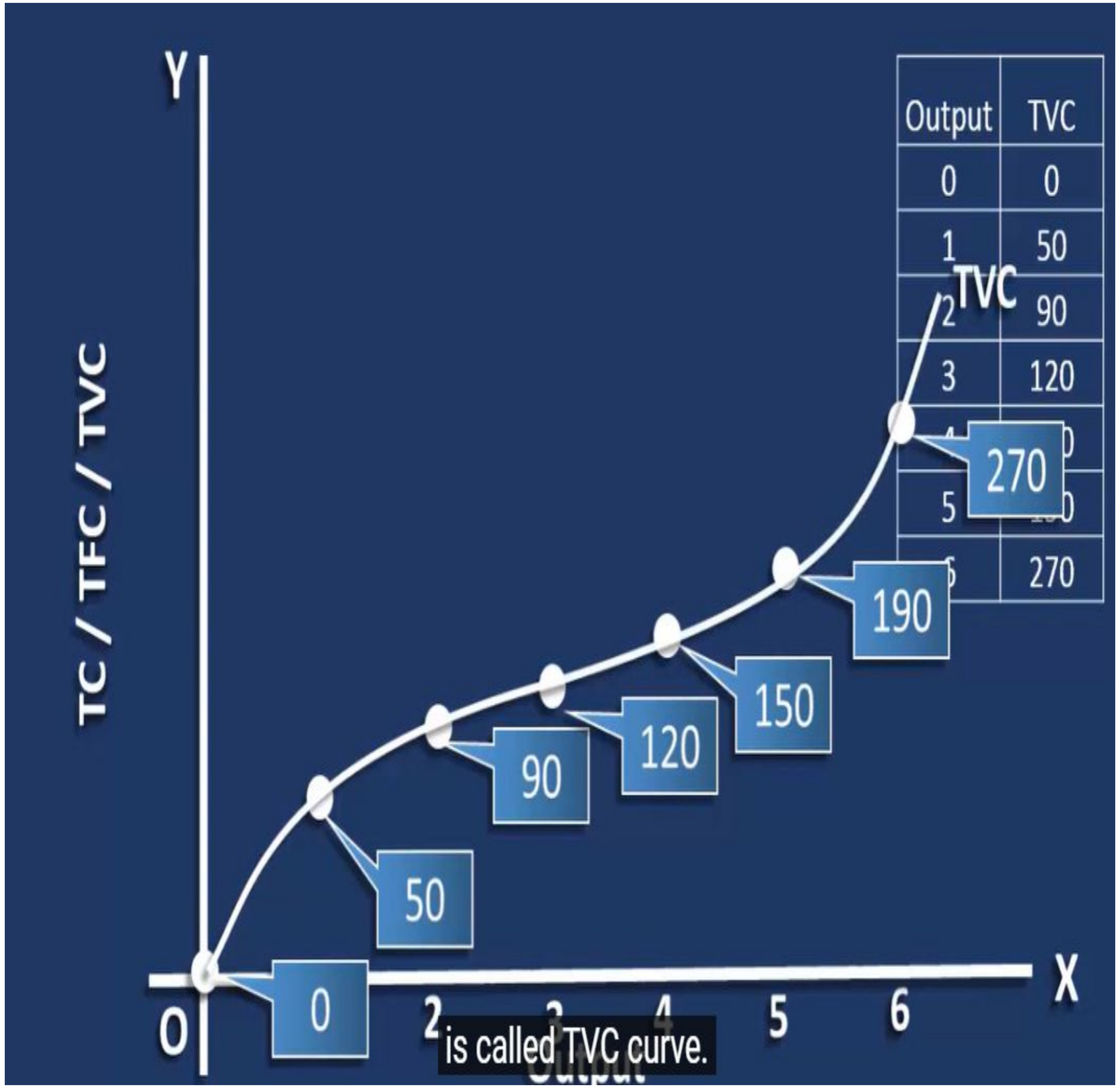
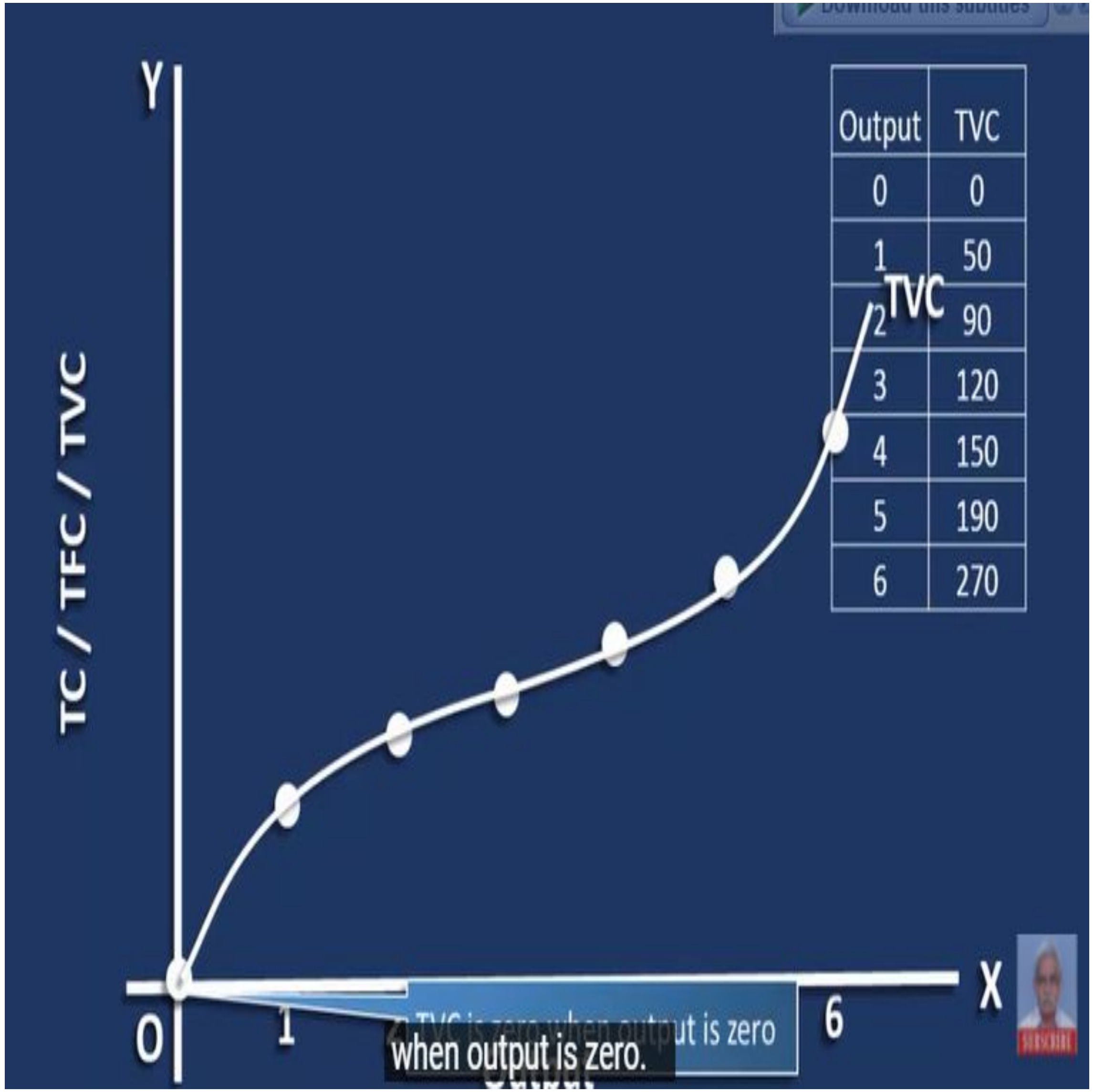
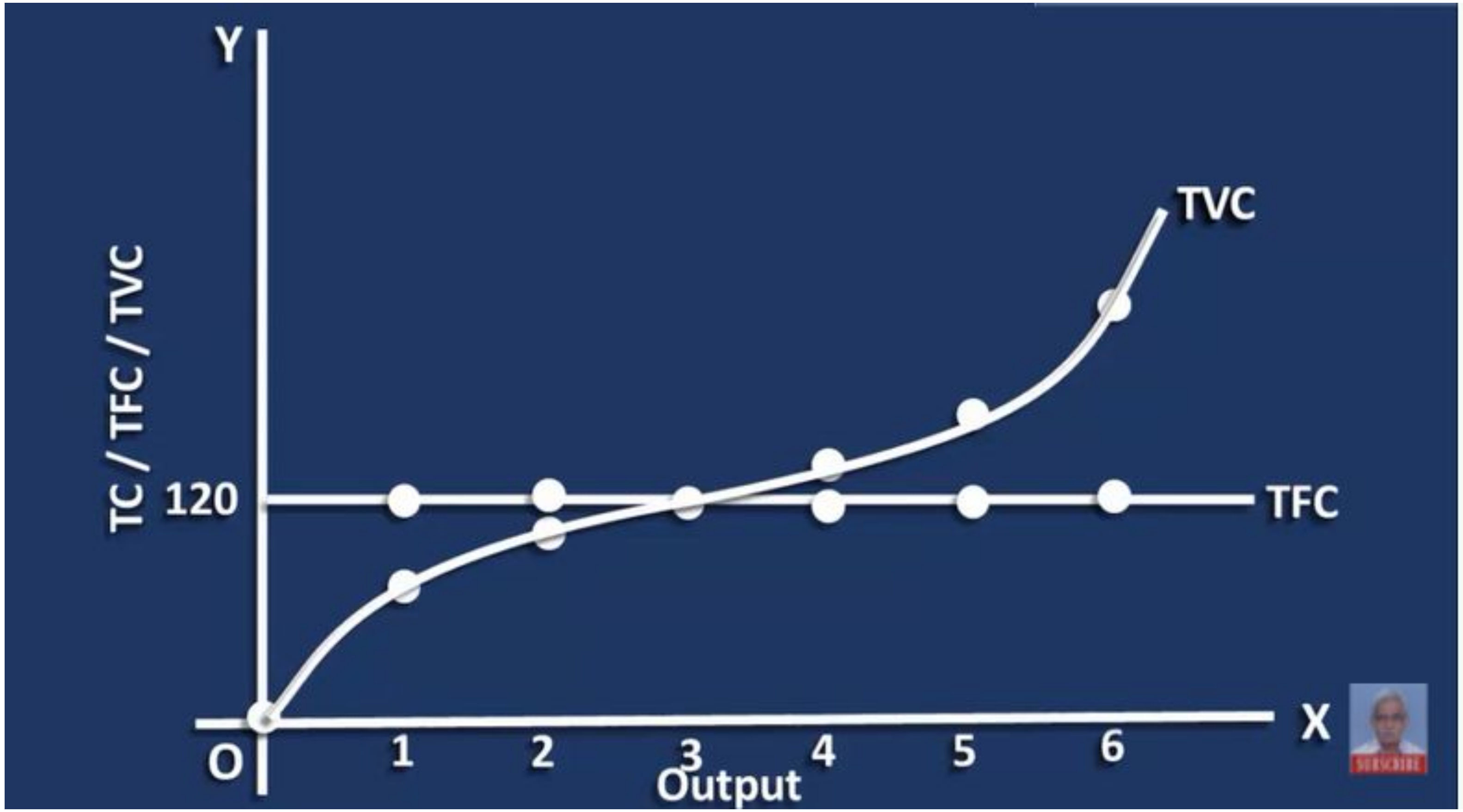
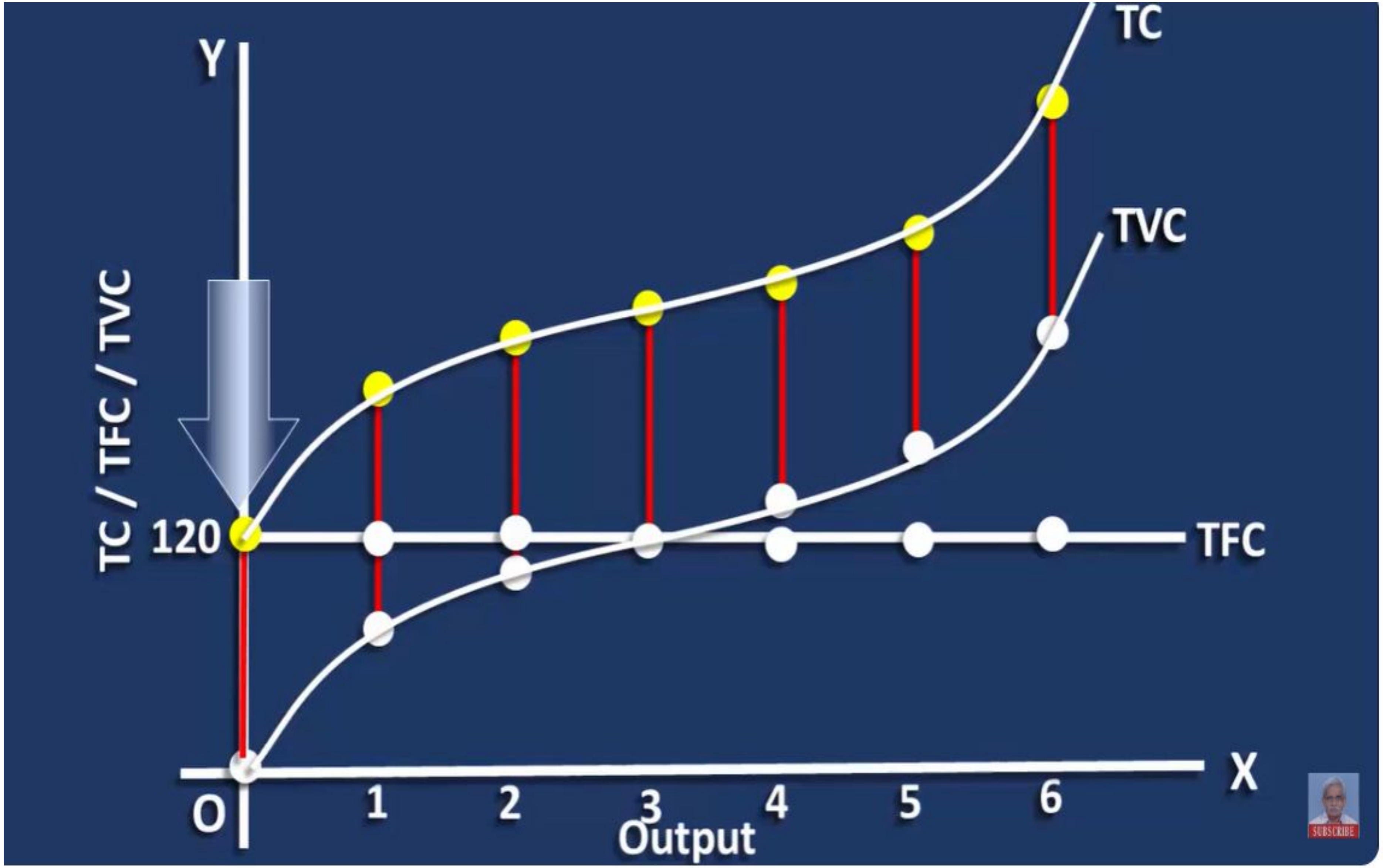

Y
1. It is parallel to x-axis.
2. Fixed cost is there, even at zero output.
3. Fixed cost doesn't change with output.
Output TFC
0 120
1 120
2 120
3 120
4 120
5 120
6 120
TC / TFC / TVC
120
TFC
X
O
1
2
3
4
5
6
Output


Y
TC / TFC / TVC
Output
TVC
0
0
1
50
2
90
3
120
TVC
270
190
150
120
90
50
0
X
O
5
6
is called TVC curve.


Y
TC / TFC / TVC
Output
TVC
0
0
1
50
2
90
3
120
4
150
5
190
6
270
TVC
X
O
1
6
when output is zero.


Y
TC / TFC / TVC
120
TVC
TFC
X
O
1
2
3
4
5
6
Output



Y
TC
TVC
TC / TFC / TVC
120
TFC
O
1
2
3
4
5
6
X
Output
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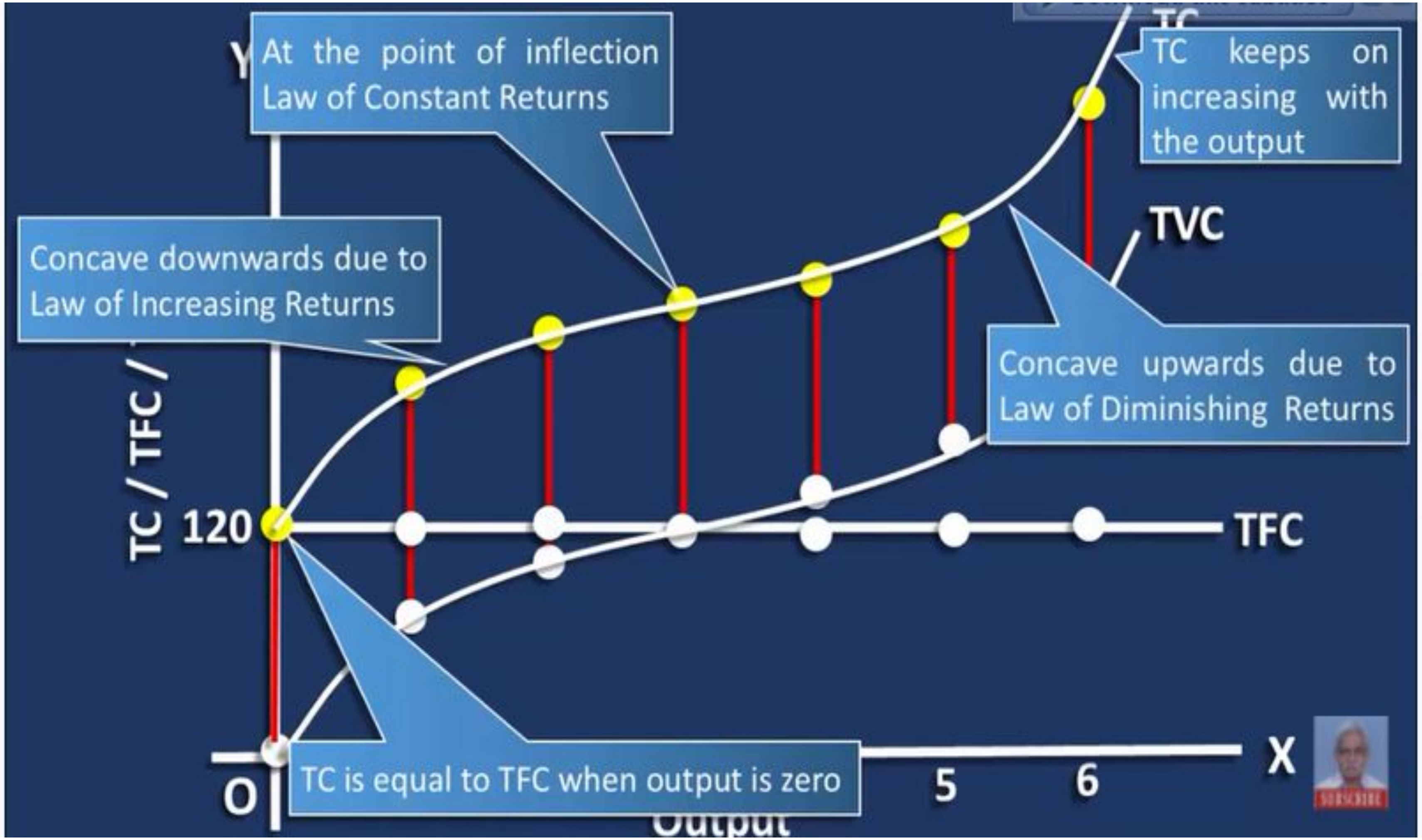


At the point of inflection Law of Constant Returns
TC keeps on increasing with the output
Y
TC
TVC
Concave downwards due to Law of Increasing Returns
Concave upwards due to Law of Diminishing Returns
TC / TFC /
120
TFC
X
O
TC is equal to TFC when output is zero
5
6

AC, AFC, AVC, MC

$$AFC = \frac{TFC}{TQ}$$

$$AVC = \frac{TVC}{TQ}$$

$$MC = \frac{\Delta TC}{\Delta TQ}$$

$$AC = \frac{TC}{TQ}$$

$$= \frac{TFC + TVC}{TQ}$$

$$= AFC + AVC$$

Table 4.8: Behaviour of average cost

Output (Units)	TFC (Birr)	TVC (Birr)	TC (Birr)	AFC (Birr)	AVC (Birr)	ATC (Birr)	MC (Birr) = (ΔTC/ΔQ)
(1)	(2)	(3)	(4) (2 + 3)	(5) (2÷1)	(6) (3÷1)	(7) (4÷1) or (5+6)	(8) ($TC_n - TC_{n-1}$)
0	60	0	60	–	–	–	–
1	60	40	100	60	40	100	40
2	60	76	136	30	38	68	36
3	60	102	162	20	34	54	26
4	60	132	192	15	33	48	30
5	60	170	230	12	34	46	38
6	60	222	282	10	37	47	52

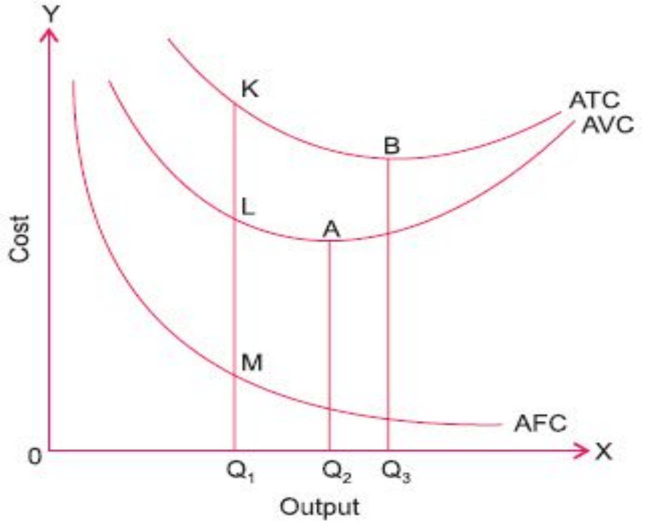


Figure 4.11: Behaviour of Average Costs

Marginal Cost

- *Marginal cost is the addition to total cost as one more unit of output is produced.*

- In other words, marginal cost is the addition to total cost of producing *n* units instead of *n* – 1 units.

$$MC_n = TC_n - TC_{n-1}$$

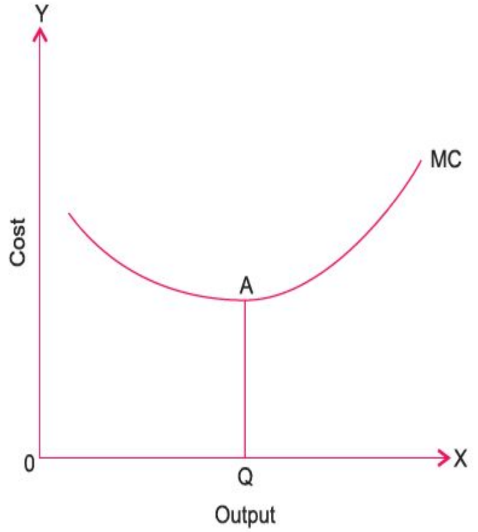
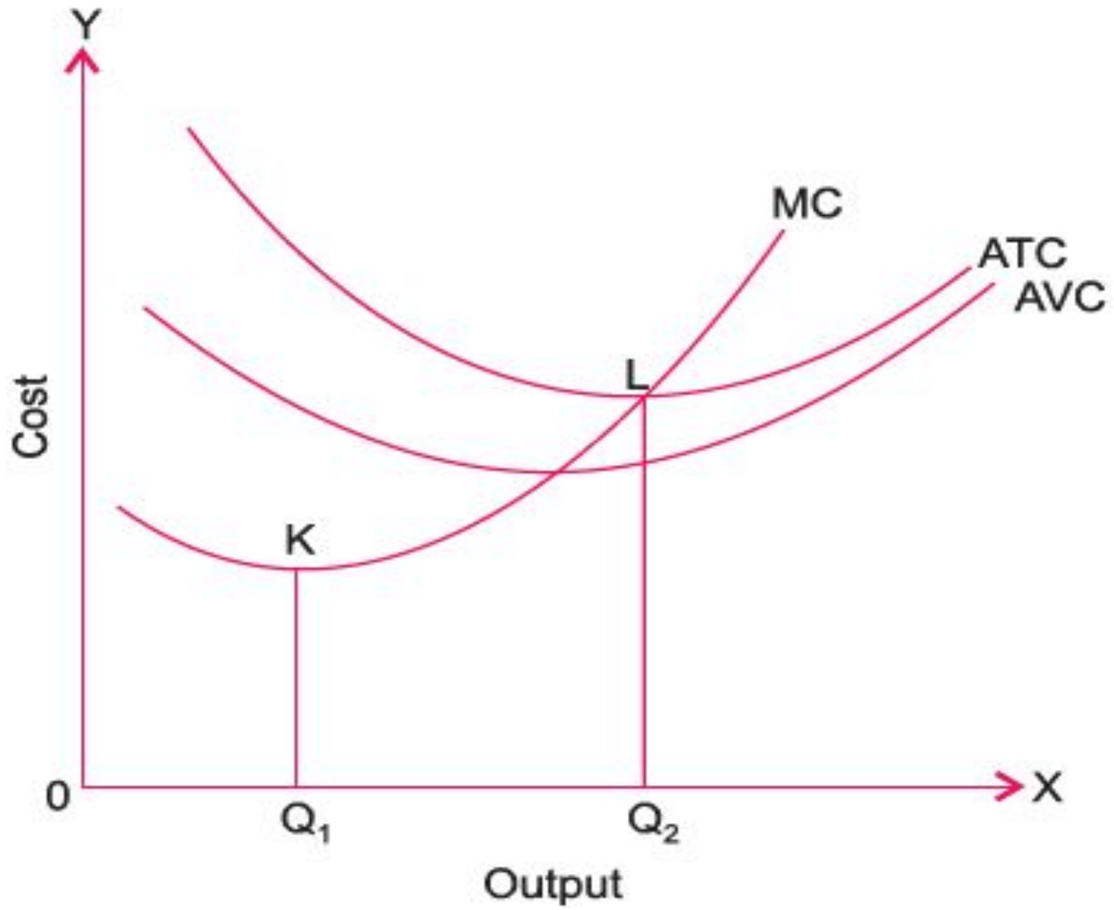


Figure 4.12: Short-Run MC Curve

The following points should be noted with regard to marginal cost:

- *Marginal cost has nothing to do with fixed cost. It is associated with the variable cost and thereby with total cost.*
- *The MC curve is U-shaped. As output increases, the MC curve slopes downward (up to OQ units), them reaches the minimum (at point A) and then starts sloping upward beyond the OQ level of output. The U-shape of the MC curve is because of the law of variable proportions. It is negatively sloped in the initial stage of production due to increasing returns to a factor, and it is positively sloped thereafter due to decreasing returns.*

Relationship Among MC, AC & AVC



Relation between Production & Cost

Table 4.9: Production and Cost in the Short Run

Units of Variable Input, Labour (L)	Total Product (Q)	Total Variable Cost (L × wage rate of Birr 100)	Marginal Cost (Birr) (ΔTVC/ΔQ)	Marginal Product (ΔQ/ΔL)
0	0	0	—	—
1	10	100	10.00	10
2	22	200	8.33	12
3	40	300	5.55	18
4	55	400	6.67	15
5	62	500	14.33	7
6	65	600	33.33	3
7	60	700	(–)20.00	(–)5

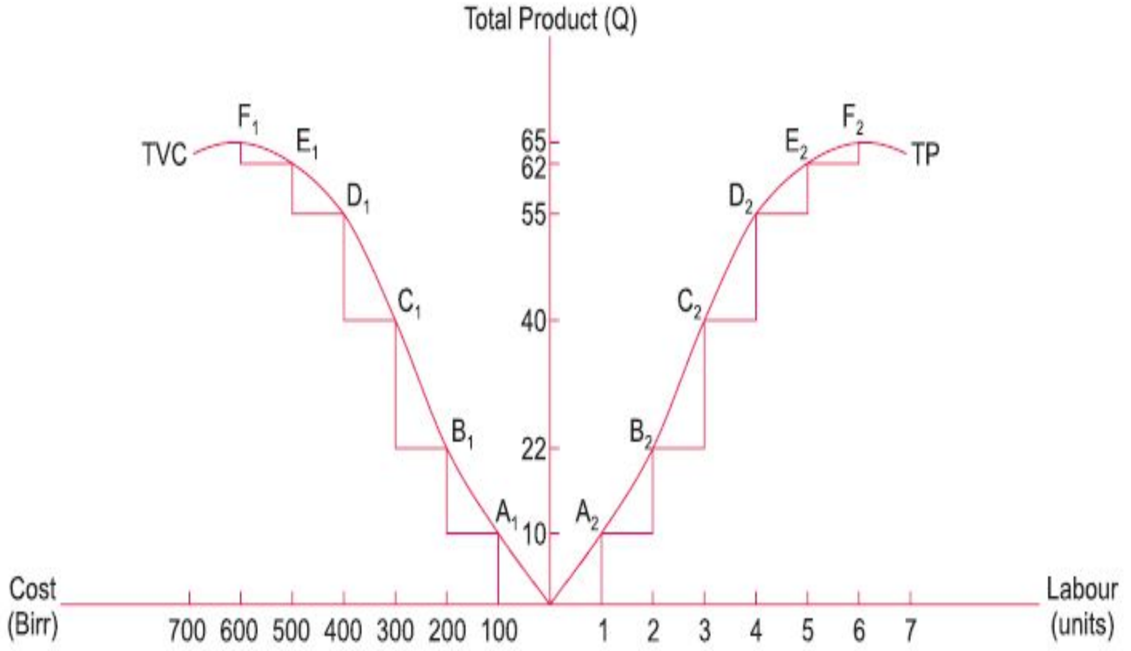


Figure 4.14: Cost and Production Short Run

Relationship between Production and Cost

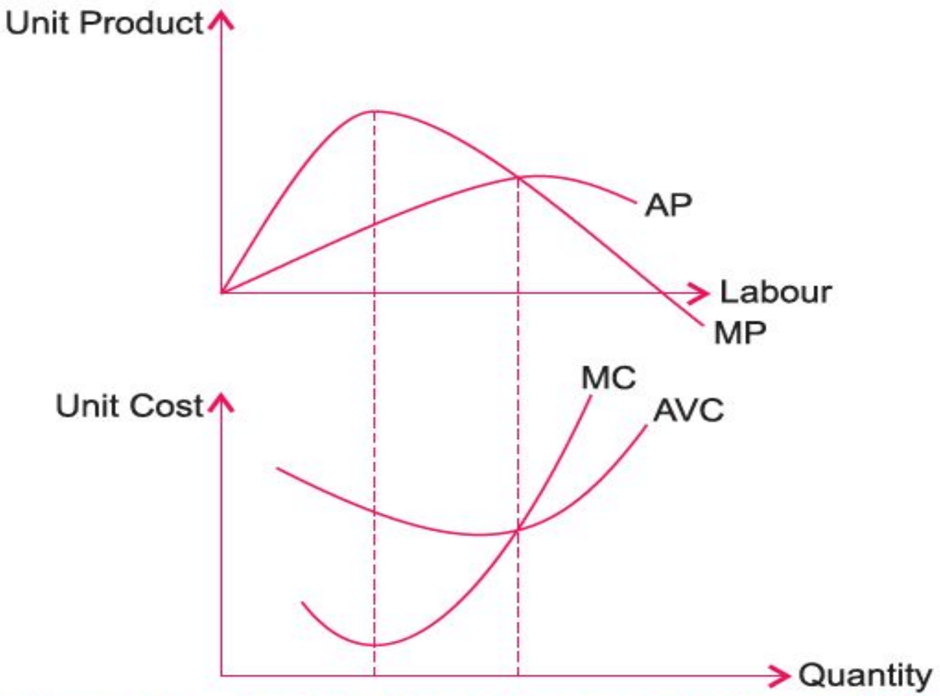


Figure 4.15: Relationship Between Production and Cost

Some other costs

- Private: *Private cost refers to cost of production incurred by an individual firm in*
- *producing a commodity.*
- **Social Cost:** *Social cost,* on the other hand, *refers to the cost that the society has to bear on account of production of a commodity.*

Private cost is the sum total of the cost incurred by the producers of goods and services (private cost) and the cost experienced bythose who have to suffer be cause of the production of the commodity in terms of external cost.

Social Cost = Private Cost + External Cost

- ***Explicit costs****: Actual payments made by a firm for purchasing or hiring resources* (or factor-services) *from the factor-owners or other firms are called explicit costs*.
- ***Implicit costs*** *refer to the imputed costs of the factors of production owned by the producer himself/herself*. They are called implicit costs because producers do not make payment to others for them. For instance, rent of his/her own land, interest on his/her own capital, and salary for his/her own services as manager, etc. are implicit costs.
- **Economic Cost = Explicit Cost + Implicit Cost**

Concepts of Total, Average, and Marginal Revenue

- **Total revenue** is the income the firm generates from selling its products. We calculate it by multiplying the price of the product times the quantity of output sold:

  Total Revenue = Price × Quantity

- Marginal revenue is the change in total revenue when one more unit of a commodity is sold.

  MR= change in TR/change in quantity sold

- Average revenue refers to revenue per unit of output.

  AR=TR/Q

- Each business, regardless of size or complexity, tries to earn a profit:

  Profit = Total Revenue – Total Cost

A Numerical Example Table 8.6 presents some data for another hypothetical firm. Let us assume that the market has set a $15 unit price for the firm's product. Total revenue in column 6 is the simple product of $P \times q$ (the numbers in column 1 times $15). The table derives total, marginal, and average costs exactly as Table 8.4 did. Here, however, we have included revenues, and we can calculate the profit, which is shown in column 8.

TABLE 8.6 Profit Analysis for a Simple Firm

(1) q	(2) TFC	(3) TVC	(4) MC	(5) $P = MR$	(6) TR $(P \times q)$	(7) TC $(TFC + TVC)$	(8) Profit $(TR - TC)$
0	$10	$ 0	$–	$15	$ 0	$10	$–10
1	10	10	10	15	15	20	–5
2	10	15	5	15	30	25	5
3	10	20	5	15	45	30	15
4	10	30	10	15	60	40	20
5	10	50	20	15	75	60	15
6	10	80	30	15	90	90	0

ECONOMIES OF SCALE

- ***Small- and large-scale Production:***
- Small scale production is the production with small capital outlay and therefore at a low level of output.
- On the other hand, large scale production is production with a large-scale outlay and therefore results in high level of output.

Economies of scale refer to the cost advantages that businesses experience as they increase their production scale. These advantages result in a reduction in the average cost per unit of production due to operational efficiencies and the spreading of fixed costs over a larger output.

- ***Economies of large-scale production*** are classified by **Marshall** into:
-
- (1) Internal Economies and (2) External Economies.

1. *Internal Economies of scale:*

- **Internal economies of scale** are those economies which are internal to the firm. These arise within the firm as a result of increasing the scale of output of the firm.

- (i) **Technical Economies**: When production is carried on a large scale, a firm can afford to install up to date and costly machinery and can have its own repairing arrangements. As the cost of machinery will be spread over a very large volume of output, the cost of production per unit will therefore, be low.
-
- A large establishment can utilize its byproducts. This will further enable the firm to lower the price per unit of the main product. A large firm can also secure the services of experienced entrepreneurs and workers which a small firm cannot afford. In a large establishment there is much scope for specialization of work, so the division of labor can be easily secured.
-
- (ii) **Managerial Economies**: When production is carried on a large scale, the task of manager can be split up into different departments and each department can be placed under the supervision of a specialist of that branch. The difficult task can be taken up by the entrepreneur himself. Due to this functional specialization, the total return can be increased at a lower cost.
-

Internal Economies of scale:

- (iii) **Marketing Economies**: Marketing economies refer to those economies which a firm can secure from the purchase or sale of the commodities. A large establishment is in a better position to buy the raw material at a cheaper rate because it can buy that commodity on a large scale. At the time of selling the produced goods, the firm can secure better rates by effectively advertising in the newspapers, journals and radio, etc.
-
- (iv**) Financial Economies:** Financial economies arise from the fact that a big establishment can raise loans at a lower rate of interest than a small establishment which enjoys little reputation in the capital market.
-
- (v) **Risk Bearing Economies:** A big firm can undertake risk bearing economies by spreading the risk. In certain cases, the risk is eliminated altogether. A big establishment produces a variety of goods in order to cater the needs of different tastes of people. If the demand for a certain type of commodities slackens, it is counter balanced by the increase in demand of the other type of commodities produced by the firm.

2. External Economies of Scale

•***External economies of scale*** are those economies which are not specially availed of by any firm. Rather these accrue to all the firms in an industry as the industry expands. The main external economies are as under:

•**(i) Economies of localization.** When an industry is concentrated in a particular area, all the firms situated in that locality avail of some common economies such as (a) skilled labor, (b) transportation facilities, (c) post and telegraph facilities, (d) banking and insurance facilities etc.

•**(ii) Economies of vertical disintegration.** The vertical disintegration implies the splitting up the production process in such a manner that some Job are assigned to specialized firms. For example, when an industry expands, the repair work of the various parts of the machinery is taken up by the various firms' specialists in repairs.

•**(iii) Economies of information.** As the industry expands it can set up research institutes. The research institutes provide market information, technical information etc. for the benefit of all the firms in the industry.

•**(iv) Economies of byproducts.** All the firms can lower the costs of production by making use of waste materials.

Internal Diseconomies of scale:

- The main ***factors causing diseconomies*** of scale and eventually leading to higher per units' cost are as follows:

- **(i) Lack of co-ordination.** As a firm becomes large scale producer, it faces difficulty in coordinating the various departments of production. The lack of co-ordination in the production, planning, marketing personnel, account, etc., lowers efficiency of the factors of production. The average cost of production begins to rise.

- **(ii) Loose control.** As the size of plant increases, the management loses control over the productive activities. The misuse of delegation of authority, the redtapisim bring diseconomies and lead to higher average cost of production.

- **(iii) Lack of proper communication.** The lack of proper communication between top management and the supervisory staff and little feedback from subordinate staff causes diseconomies of scale and results in the average cost to go up.
-

- **(iv) Lack of identification.** In a large organizational structure, there is no close liaison between the top management and the thousands of workers employed in the firm. The lack of identification of interest with the firm results in the per unit cost to go up

External Diseconomies of scale:

- A firm or an industry cannot avail of economies for an indefinite period of time. With the expansion and growth of an industry, certain disadvantage also begins to arise. The ***diseconomies of large-scale production*** are:
-
- (i) Diseconomies of pollution, (ii) Excessive pressure on transport facilities, (iii) Rise in the prices of the factors of production, (iv) Scarcity of funds, (v) Marketing problems of the products, (iv) Increase in risks.
-